

▷ RISING PEAKS

As the African Plate drives north, it has crushed southern Europe and thrown up the Alps and other high mountain ranges such as the Carpathians and Caucasus.

▷ DEEP VALLEY

A collision between tectonic plates 400 million years ago opened a gash across the Scottish Highlands called the Great Glen—which is now filled by Loch Ness.



THE SHAPING OF EUROPE

Europe is a relatively small continent, yet has its own unique geology. Whereas most continents are formed from several stable sections of continental crust called cratons, Europe was forged as fragments formed and reformed around just one—the East European Craton.

Europe rises

The East European Craton lies buried under sediments across the North European Plain that stretch from the Netherlands to the Ural Mountains. But in Sweden and Finland, the Craton is exposed.

At first, the East European Craton formed the continent of Baltica. Then, 430 million years ago, Baltica collided with ancient landmasses geologists call Laurentia (now North America) and Avalonia (now the British Isles and part of North America) to create the supercontinent Euramerica. Along the collision zone, mountains were thrown up across Norway and Britain—the Scottish Highlands are remnants of this range. Much of Euramerica was a desert, and it is sometimes known as the Old Red Continent for the red sandstones that formed there.

In time, Euramerica crashed into the southern supercontinent of Gondwana to help form the giant supercontinent of Pangaea, piling up new mountains, such as the Vosges and the Harz, which have

now worn down. Pangaea eventually broke up to form the continents we know today, with North America splitting away from Europe as the Atlantic Ocean opened up. But the separated continent of Africa has for the last 40 million years been driving relentlessly north, forcing the southern edge of Europe up into a series of new mountain ranges from the Pyrenees to the Alps and the Caucasus.

Deep freeze

For much of the last two million years, in extremely cold periods known as ice ages, vast ice sheets covered northern Europe. Their grinding movement scoured large areas of lowland Europe flat, and left behind extensive deposits of debris, known as moraines, creating a bumpy landscape. In Scotland and the Lake District, glaciers carved deep, troughlike valleys. In Norway, they gouged out

5.6 million years ago,
the Mediterranean Basin
was a desert

KEY EVENTS

70 million years ago The Mid-Atlantic Ridge is spreading. In one area, hot magma from Earth's mantle cools into new crust, creating Iceland.

5.3 million years ago Tectonic activity lowers the Strait of Gibraltar to below sea level, causing the Atlantic Ocean to pour into the Mediterranean Basin, forming the Mediterranean Sea.

18,000–11,000 years ago During the coldest periods of the last ice age, glaciers carve out many of Europe's most dramatic landscapes.

10,000 years ago Vast forests start to cover Europe, replacing the grassy steppes of the ice age; remnants such as the Białowieża Forest survive today.

90 million years ago Tectonic activity in what will be southern Europe begins and will continue to eventually form the Alps.

30–20 million years ago Europe starts to form almost recognizable landmasses; most of western Europe is now above land, and a shallow ocean that separates Europe from Asia disappears.

9,000–8,000 years ago A tsunami floods the land link between what will become the British Isles and mainland Europe, forming the English Channel.